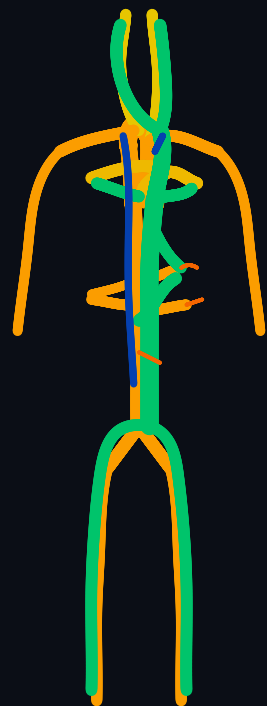


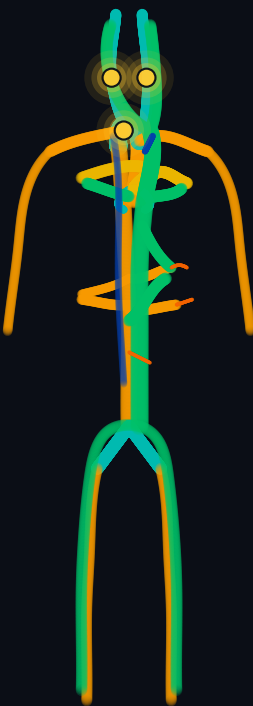
Wall Shear Stress Across Disease Scenarios

Healthy baseline



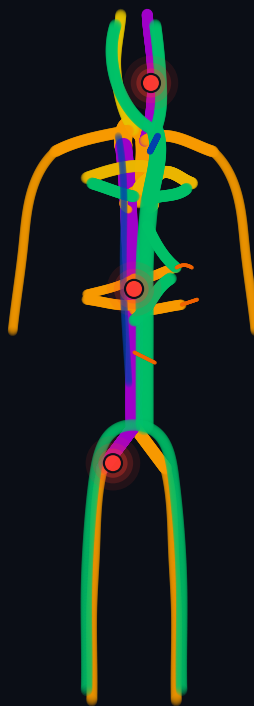
Across a healthy circulation, wall shear stress spans four orders of magnitude —

Atherosclerosis



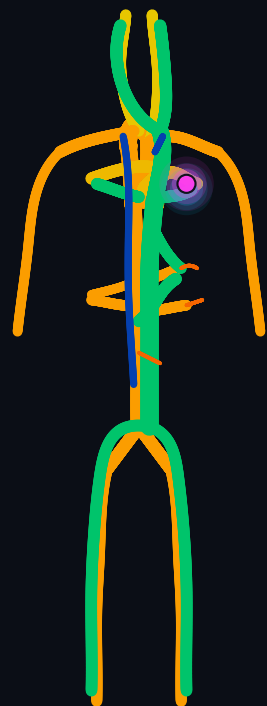
Plaque forms where shear is low and oscillatory (below ~ 4 dyne/cm²) — typically at bifurcations.

Arterial stenosis



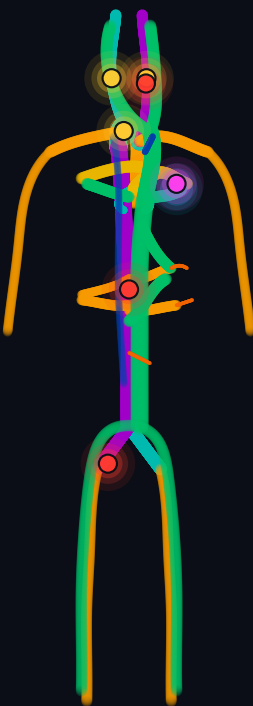
Severe narrowing creates shear hotspots above 1000 dyne/cm² — extreme stress that can strip a carrier's hydration shell and...

Tumor vasculature



Tumor microvasculature is disorganised, with low and oscillatory shear and elevated interstitial fluid pressure —

Combined pathology



Co-existing atherosclerosis, stenosis and tumor vasculature in one patient —

Hotspot legend

- Plaque (atherosclerosis)
- Stenosis hotspot
- Tumor vasculature

Affected vessels recoloured by representative WSS; unaffected vessels dimmed for context.



Wall shear stress (dyne/cm², log scale)